Rajat Savdekar

rajatsavdekar@gmail.com • linkedin.com/in/rajatsavdekar • rajatsavdekar.dev • (213)-574-8894 • Los Angeles, CA

Work Authorization: F-1 OPT (3 Years STEM Eligible till May 2029) • No Immediate Sponsorship Required • Available Immediately

EDUCATION

University of Southern California — Los Angeles, CA
Master of Science in Computer Science (GPA: 3.63) — *Aug 2024 – May 2026*

University of Mumbai — Mumbai, India
Bachelor of Engineering in Computer Engineering (CGPA: 9.4/10) — *Aug 2019 – May 2023*

SKILLS

Languages: Python, TypeScript, JavaScript, C++, C#, Java, SQL, React, Next.js, HTML, Tailwind CSS
Backend & Distributed Systems: FastAPI, Flask, Node.js, REST APIs, WebSockets, gRPC, Microservices, Redis
Cloud & Infrastructure: AWS (EC2, S3, CloudWatch), Docker, PostgreSQL, Terraform, GitHub Actions, CI/CD

EXPERIENCE

Easley-Dunn Productions, Inc. – Mediverse — Torrance, CA
Software Developer — *Aug 2026 – Present*

- Cut replay pipeline latency from 22 minutes to 15 seconds by engineering an asynchronous **FastAPI** job runner and **Django REST** endpoints with **React** state hooks that memoize event state and stream telemetry data.
- Reduced frame-time spikes by 30% through low-level **C#** and memory optimizations across surgical simulation subsystems, eliminating garbage collection stalls via object pooling.
- Compressed on-device LLM runtime memory to under 4 GB RAM by applying **4-bit AWQ quantization** and KV-cache optimizations for real-time low-latency inference.

Jalgaon Fruit Sales Cooperative Ltd (JFSS) – AgriTech Supply Chain ERP — Jalgaon, India
Software Engineer II — *May 2023 – May 2024*

- Doubled peak harvest write throughput by decomposing a **legacy PHP monolith** into **Python Flask microservices** on **PostgreSQL**, eliminating MyISAM table locks and cutting query latency 40%.
- Architected a semantic vector search service over 150K+ procurement records using **ChromaDB** and **OpenAI embeddings**, cutting average document retrieval latency from 3 minutes to under 20 seconds.
- Deployed containerized services with **Docker** to **AWS EC2** via **GitHub Actions** CI/CD pipelines monitored with CloudWatch alerts.
- Engineered an asynchronous reconciliation pipeline for 1,000+ daily invoices using **Redis queues** with exponential retry backoff, isolating third-party banking rate limits.

Software Engineer I — *May 2022 – May 2023*

- Architected idempotent transactional **REST APIs** in **Python Flask** with **SQLAlchemy ORM** to ingest real-time weighbridge scale metrics and calculate multi-tier farmer disbursement payouts under ACID guarantees.
- Built distributed batch reconciliation workers with **Celery**, **Redis**, and **cron** to parse multi-bank statement streams, executing double-entry ledger settlement against internal journal entries to eliminate reconciliation drift.
- Reduced slow-query latency spikes by **35%** across high-volume harvest ledgers by designing B-tree composite indices, rewriting nested subqueries into indexed **JOINs**, and tuning PostgreSQL query execution plans.

Softaid Computers — Jalgaon, India
Software Engineering Intern — *Jul 2021 – Dec 2021*

- Automated shift scheduling and multi-branch ledger aggregation by developing **RESTful** services in **JavaScript** and **MySQL** for the **E-Attendance** and **E-Vikas** modules, cutting administrative reporting lag by 65%.

PROJECTS

MARA – Multi-Agent Research Intelligence Platform [GitHub]
Python, LangGraph, FastAPI, ChromaDB, Docker, WebSockets — *Sep 2026*

- Built a multi-agent orchestration engine using **LangGraph** and **FastAPI** that coordinates specialized planner, web-scraping, and synthesis agents via stateful directed graphs to generate comprehensive research dossiers.
- Streamed agent execution traces and intermediate outputs in real time via **WebSockets**, integrating semantic vector search across retrieved citations using **ChromaDB** to eliminate cross-agent hallucinations.

MockPad – Real-Time Collaborative Code Editor [Live] [GitHub]
TypeScript, React, Node.js, WebSockets, MongoDB — *May 2026*

- Scaled collaborative editing to **10–20 DAU** on a $6/month server using **Node.js** and **WebSockets** at 100% uptime.
- Built an interactive interview UI with deterministic session playback in **React** and **TypeScript**.

RESEARCH & HACKATHONS

Publications. *Texture Feature Analysis* – MATLAB tool cited by 10+ papers in IEEE and Springer. [GitHub]
Hackathons. Competed in 6+ events including **OpenAI Build Week**, **CALL-E 2026**, and **Reddit Devvit**. [Demo]